

PRACTICE SHEET

1. If the matrix $\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is singular, then what is one of the values of θ ?
 (a) $\pi/4$ (b) $\pi/2$
 (c) π (d) 0
2. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix}$, If $AB = BA$, then what is the value of x ?
 (a) -1 (b) 0
 (c) 1 (d) Any real number
3. Let $A = (a_{ij})_{n \times n}$ and $\text{adj}(A) = (\alpha_{ij})$, If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 2 & 3 & -1 \end{bmatrix}$ then what is the value of α_{23} ?
 (a) 1 (b) -1
 (c) 8 (d) -8
4. Let A and B be two invertible matrices each of order n . What is $\text{adj}(AB)$ equal to?
 (a) $(\text{adj}A)(\text{adj}B)$ (b) $(\text{adj}A) + (\text{adj}B)$
 (c) $(\text{adj}A) - (\text{adj}B)$ (d) $(\text{adj}B)(\text{adj}A)$
5. M is a matrix with real entries given by $M = \begin{bmatrix} 4 & k & 0 \\ 6 & 3 & 0 \\ 2 & t & k \end{bmatrix}$, which of the following conditions guarantee the invertibility of M ?
 1. $k \neq 2$ 2. $k \neq 1$
 3. $t \neq 0$ 4. $t \neq 1$
 Select the correct answer using the code given below:
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 4 (d) 3 and 4
6. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ be a square matrix of order 3. Then for any positive integer n , what is A^n equal to?
 (a) A (b) $3^n A$
 (c) $(3^{n-1}) A$ (d) $3A$
7. If A is a square matrix such that $A - A^T = 0$, then which one of the following is correct?
 (a) A must be a null matrix
 (b) A must be a unit matrix
 (c) A must be a scalar matrix
 (d) None of the above
8. What is the inverse of $A = \begin{bmatrix} 1+i & 1+i \\ -1+i & 1-i \end{bmatrix}$?
 (a) $\frac{1}{4} \begin{bmatrix} 1-i & -1-i \\ 1-i & 1-i \end{bmatrix}$ (b) $\frac{1}{4} \begin{bmatrix} 1+i & -1+i \\ 1+i & -1-i \end{bmatrix}$
 (c) $\frac{1}{4} \begin{bmatrix} 1+i & 1-i \\ -1-i & 1+i \end{bmatrix}$ (d) $\frac{1}{4} \begin{bmatrix} 1+i & 1-i \\ -1-i & -1+i \end{bmatrix}$
9. Let A be an $m \times n$ matrix. Under which one of the following conditions does A^{-1} exist?
 (a) $m = n$ only
 (b) $m = n$ and $\det(A) \neq 0$
 (c) $m = n$ and $\det(A) = 0$
 (d) $m \neq n$
10. Let A and B be two matrices of order $n \times n$. Let A be a non-singular. Consider the following:
 (a) AB is a singular
 (b) AB is a non-singular
 (c) $A^{-1}B$ is a singular
 (d) $A^{-1}B$ is non-singular
11. The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ satisfies which one of the following polynomial equations?
 (a) $A^2 + 3A + 2I = 0$ (b) $A^2 + 3A - 2I = 0$
 (c) $A^2 - 3A - 2I = 0$ (d) $A^2 - 3A + 2I = 0$
12. If a matrix A is symmetric as well as anti-symmetric, then which one of the following is correct?
 (a) A is a diagonal matrix
 (b) A is a null matrix
 (c) A is a unit matrix
 (d) A is a triangular matrix
13. If $A = \begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$, where ω is cube root of unity, then what is A^{100} equal to:
 (a) A (b) $-A$
 (c) Null matrix (d) Identity matrix
14. If a matrix X has $(a+b)$ rows and $(a+2)$ columns and a matrix Y has $(b+1)$ rows and $(a+3)$ columns. If both XY and YX exist, then what are the values of a, b respectively?
 (a) 3, 2 (b) 2, 3
 (c) 2, 4 (d) 4, 3
15. If X and Y are the matrices of order 2×2 each and $2x - 3y = \begin{bmatrix} -7 & 0 \\ 7 & -13 \end{bmatrix}$ and $3x + 2y = \begin{bmatrix} 9 & 13 \\ 4 & 13 \end{bmatrix}$, then what is Y equal to:
 (a) $\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$
 (c) $\begin{bmatrix} 3 & 2 \\ -1 & 5 \end{bmatrix}$ (d) $\begin{bmatrix} 3 & 2 \\ 1 & -5 \end{bmatrix}$
16. If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, then $A \cdot \text{adj} A$ is equal to:
 (a) $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 10 \\ 10 & 0 \end{bmatrix}$
 (c) $\begin{bmatrix} 10 & 10 \\ 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 10 & 10 \end{bmatrix}$

17. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ where a, b are natural numbers, then which one of the following is correct?
 (a) There exist more than one but finite number of B 's such that $AB=BA$
 (b) There exists exactly one B such that $AB = BA$
 (c) There exists infinitely B 's such that $AB = BA$
 (d) There cannot exist any B such that $AB = BA$
18. What is the inverse of $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$?
- (a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
 (c) $\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$
19. Consider the following statements in respect of symmetric matrices A and B .
 I. AB is symmetric
 II. $A^2 + B^2$ is symmetric
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
20. Which one of the following is correct in respect of the matrix?
 $A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$?
- (a) A^{-1} does not exist (b) $A = (-1)I$
 (c) A is a unit matrix (d) $A^2 = I$
21. Let A and B matrices of order 3×3 . If $AB = 0$, then which of the following can be concluded?
 (a) $A = 0$ and $B = 0$
 (b) $|A|=0$ and $|B|=0$
 (c) Either $|A|=0$ or $|B|=0$
 (d) Either $A = 0$ and $B = 0$
22. Consider the following statements in respect of a square matrix A and its transpose A^T .
 I. $A + A^T$ is always symmetric
 II. $A - A^T$ is always anti-symmetric

- Which of the statements given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
23. If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, $C = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$, then which one of the following is not correct?
 (a) $A^2 = B^2$ (b) $B^2 = C^2$
 (c) $AB = C$ (d) $AB = BA$
24. If A is a square matrix, then what is $\text{adj}(A^T) - (\text{adj } A)^T$ equal to?
 (a) $2|A|$ (b) $2|A|I$
 (c) Null matrix (d) Unit matrix
25. If a matrix A is such that $3A^3 + 2A^2 + 5A + I = 0$, then A^{-1} is equal to:
 (a) $-(3A^2 + 2A + 5)$ (b) $3A^2 + 2A + 5$
 (c) $3A^2 - 2A - 5$ (d) None of these
26. Let $A = \begin{bmatrix} 5 & 6 & 1 \\ 2 & -1 & 5 \end{bmatrix}$, if there exists a matrix B such that $AB = \begin{bmatrix} 35 & 49 \\ 29 & 13 \end{bmatrix}$, then what is B equal to?
 (a) $\begin{bmatrix} 5 & 1 & 4 \\ 2 & 6 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 6 & 3 \\ 5 & 1 & 4 \end{bmatrix}$
 (c) $\begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 5 \\ 6 & 1 \\ 3 & 4 \end{bmatrix}$
27. Consider the following statements:
 I. If $A' = A$, then A is singular matrix, where A' is the transpose of A .
 II. If A is a square matrix such that $A^3 = I$, then A is non-singular.
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
28. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $[a_{ij}]$, where, $i, j = 1, 2$. If its inverse matrix is $[b_{ij}]$, then what is b_{22} equal to?
 (a) -2 (b) 1
 (c) $1/3$ (d) $-1/2$

ANSWER KEY

1.	a	2.	b	3.	c	4.	d	5.	a	6.	c	7.	d	8.	a	9.	b	10.	b
11.	C	12.	b	13.	a	14.	b	15.	c	16.	a	17.	c	18.	b	19.	b	20.	d
21.	c	22.	c	23.	b	24.	c	25.	a	26.	c	27.	b	28.	d				

Solutions

Sol.1. (a)

$$\begin{vmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = 0$$

$$\cos^2 \theta - \sin^2 \theta = 0$$

$$\cos 2\theta = 0$$

$$\theta = \frac{\pi}{4}$$

Sol.2. (b)

$$AB = BA$$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & x \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -x \\ 0 & -1 \end{bmatrix} \Rightarrow x = 0$$

Sol.3. (c)

$$\alpha_{23} = a_{32} = (-1)^{2+3}$$

$$\begin{vmatrix} 1 & 3 \\ 4 & 4 \end{vmatrix} = -(4-12) = 8$$

Sol.4. (d)

Since A and B be two invertible square matrices each of order n then $(AB)^{-1} = (B^{-1})(A^{-1})$

$$\Rightarrow \frac{\text{adj}(AB)}{|AB|} = \frac{\text{adj}(B)}{|B|} \cdot \frac{\text{adj}(A)}{|A|}$$

Since $|AB| = |B| |A|$

then $\text{adj}(AB) = (\text{adj}(B))(\text{adj}(A))$

Sol.5. (a)

$$M = \begin{bmatrix} 4 & k & 0 \\ 6 & 3 & 0 \\ 2 & t & k \end{bmatrix} \neq 0$$

$$\Rightarrow k(12-6k) \neq 0$$

$$\Rightarrow k \neq 0 \text{ and } k \neq 2$$

Sol.6. (c)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^3 = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = 3 \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix}$$

$$= 9 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = 3^2 A$$

Hence $A^n = 3^{n-1} A$

Sol.7. (d)

$A = A^T$ answer is symmetric matrix

Sol.8. (a)

$$A = \begin{bmatrix} 1+i & 1+i \\ -1+i & 1-i \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} 1-i & -1-i \\ 1-i & 1+i \end{bmatrix} \text{ and } |A| = 4$$

$$\text{then } A^{-1} = \frac{1}{4} \begin{bmatrix} 1-i & -1-i \\ 1-i & 1-i \end{bmatrix}$$

Sol.9. (b)

Sol.10. (b)

$$|AB| = |A||B|$$

Sol.11. (c)

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 6 & 8 \end{bmatrix}$$

Let $A^2 + xA + yI = 0$ where x and y are constants.

$$A^2 + xA + yI = 0$$

$$\begin{bmatrix} 5 & 6 \\ 6 & 8 \end{bmatrix} + x \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix} + y \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 5+x+y & 6+2x \\ 6+2x & 8+2x+y \end{bmatrix} = 0$$

by solving it $x = -3$ and $y = -2$

so answer is 3rd option.

Sol.12. (b)

Since $A' = A$ and $A' = -A$

$$\Rightarrow A = -A$$

$$\Rightarrow A = O$$

Sol.13. (a)

Sol.14. (b)

The order of given matrices are

$$[X]_{(a+b) \times (a+2)} \text{ and } [Y]_{(b+1) \times (a+3)}$$

As $[XY]$ and $[YX]$ both exist

$$\therefore a+2 = b+1$$

$$\text{and } a+3 = a+b$$

$$\Rightarrow a = 2, b = 3$$

Sol.15. (c)

$$\text{Given, } 2x-3y = \begin{bmatrix} -7 & 0 \\ 7 & -13 \end{bmatrix} \dots(i)$$

$$\text{And } 3x+2y = \begin{bmatrix} 9 & 13 \\ 4 & 13 \end{bmatrix} \dots(ii)$$

On multiplying Eq. (i) by 3 and Eq. (ii) by 2 and subtracting Eq.(i) from Eq. (ii) we, get

$$13Y = 2 \begin{bmatrix} 9 & 13 \\ 4 & 13 \end{bmatrix} - 3 \begin{bmatrix} -7 & 0 \\ 7 & -13 \end{bmatrix}$$

$$\Rightarrow 13Y = \begin{bmatrix} 39 & 26 \\ -13 & 65 \end{bmatrix}$$

$$\Rightarrow Y = \begin{bmatrix} 3 & 2 \\ -1 & 5 \end{bmatrix}$$

Sol.16. (a)

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \text{ Adj}(A) = \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore A \text{ Adj}(A) = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$$

{Remember, $A \cdot \text{Adj}(A) = |A|I_n$ }

Sol.17. (c)

$$\therefore A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ and } B = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix}$$

$$\text{And } BA = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$$

If $AB = BA$, then

$$\begin{bmatrix} a & 2b \\ 3a & 4b \end{bmatrix} = \begin{bmatrix} a & 2a \\ 3b & 4b \end{bmatrix}$$

$$\Rightarrow a = b$$

From the above, it is clear that there exist infinitely many B's such that $AB = BA$.

Sol.18. (b)

$$\text{Let } A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\therefore |A| = -1 \text{ and } \text{adj}(A) = -\frac{1}{2} \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$\text{Hence, } A^{-1} = \frac{1}{|A|} \text{adj}(A) = -\frac{1}{1} \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Sol.19. (b)

Since, $(AB)' = B' A' = BA$

So, AB is not symmetric and

$$(A^2 + B^2)' = (A')^2 + (B')^2 = A^2 + B^2$$

Hence, $A^2 + B^2$ is symmetric

Sol.20. (d)

$$\therefore |A| \neq 0$$

So, A^{-1} exists

Now,

$$A^2 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Sol.21. (c)

If $AB = 0$, then it may be concluded that either $|A|=0$ or $|B|=0$,

Sol.22. (c)

Sol.23. (d)

Sol.24. (c)

$$\text{adj}(A)^T = (\text{adj } A)^T$$

Sol.25. (a)

$$\text{We have } 2A^3 + 2A^2 + 5A + I = 0$$

$$\Rightarrow I = -3A^3 - 2A^2 - 5A$$

$$\Rightarrow IA^{-1} = (-3A^2 - 2A - 5A)A^{-1}$$

$$\Rightarrow A^{-1} = -3A^2 - 2A - 5I$$

Sol.26. (c)

$$\text{By option, (c), let } B = \begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix}$$

$$AB = \begin{bmatrix} 5 & 6 & 1 \\ 2 & -1 & 5 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 1 & 6 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 35 & 49 \\ 29 & 13 \end{bmatrix}$$

Sol.27. (b)

If $A' = A$, then it is symmetric matrix

Sol.28. (d)

$$\therefore A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\therefore A^{-1} = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$\Rightarrow [b_{ij}] = -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$\Rightarrow b_{22} = -\frac{1}{2}$$

NDA PYQ

1. If the matrix $A = \begin{bmatrix} 2-x & 1 & 1 \\ 1 & 3-x & 0 \\ -1 & -3 & -x \end{bmatrix}$ is singular, then what is the solution set S?
 (a) $S = \{0, 2, 3\}$ (b) $S = \{-1, 2, 3\}$
 (c) $S = \{1, 2, 3\}$ (d) $S = \{2, 3\}$ [NDA (I) - 2011]
2. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ such that $A^2 = B$, then what is the value of α ?
 (a) -1 (b) 1
 (c) 2 (d) 4 [NDA (I) - 2011]
3. If $A = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, then which of the following is/are correct?
 I. AB is defined
 II. BA is defined
 III. $AB = BA$
 Select the correct answer using the code given below:
 (a) Only I (b) Only II
 (c) I and II (d) I, II and III [NDA (I) - 2011]
4. Consider the following statements.
 I. The inverse of the square matrix, if it exists, is unique.
 II. If A and B are singular matrices of order n , then AB is also a singular matrix of order n .
 Which of the statements given above is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II [NDA (I) - 2011]
5. If the matrix $A = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$ is such that $A^2 = I$, then which one of the following is correct?
 (a) $\alpha=0, \beta=1$ or $\alpha=1, \beta=0$ (b) $\alpha=0, \beta \neq 1$ or $\alpha \neq 1, \beta=0$
 (c) $\alpha=1, \beta \neq 0$ or $\alpha \neq 1, \beta=0$ (d) $\alpha \neq 0, \beta \neq 0$ [NDA (I) - 2011]
6. For what value of x does $(1 \ 3 \ 2) \begin{pmatrix} 1 & 3 & 0 \\ 3 & 0 & 2 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix} = (0)$ hold?
 (a) -1 (b) 1
 (c) $9/8$ (d) $-9/8$ [NDA (II) - 2011]
7. Consider the following statements.
 I. Every zero matrix is a square matrix
 II. A matrix has a numerical value
 III. A unit matrix is a diagonal matrix
 Which of the above statement is/are correct?
 (a) Only II (b) Only III
 (c) Both II and III (d) Both I and III [NDA (I) - 2012]
8. If a matrix A has inverse B and C , then which one of the following is correct?
 (a) B may not be equal to C
 (b) B should be equal to C
 (c) B and C should be unit matrices
 (d) None of the above
9. If A is a square matrix such that $A^2 = I$ where I is the identity matrix, then what is the value of A^{-1} ?
 (a) $A + I$ (b) Null matrix
 (c) A (d) Transpose of A [NDA (I) - 2012]
10. What is the order of the product:

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} ?$$

 (a) 3×1 (b) 1×1
 (c) 1×3 (d) 3×3 [NDA (I) - 2012]
11. The sum and product of matrices A and B exist. Which of the following implications are necessarily true?
 I. A and B are square matrices of same order
 II. A and B are non-singular matrices.
 Select the correct answer using the codes given below:
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II [NDA (I) - 2012]
12. If $A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$, then what is the value of $B^{-1} A^{-1}$.
 (a) $\begin{bmatrix} 1 & -3 \\ -1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$
 (c) $\begin{bmatrix} -1 & 3 \\ -1 & -2 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & -3 \\ 1 & -2 \end{bmatrix}$ [NDA (I) - 2012]
13. A and B are two matrices such that $AB=A$ and $BA=B$ then what is B^2 equal to?
 (a) B (b) A
 (c) I (d) $-I$
 Where I is the identity matrix [NDA-2012(1)]
14. If $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 & 7 \\ 4 & 6 & 8 \end{bmatrix}$, then
 Which one of the following is correct?
 (a) B is the inverse of A (b) B is the adjoint of A
 (c) B is the transpose of A (d) None of these [NDA (II) - 2012]
15. The inverse of a diagonal matrix is a:
 (a) Symmetric matrix (b) Skew-symmetric matrix
 (c) Diagonal matrix (d) None of these [NDA (II) - 2012]
16. If the matrix $\begin{bmatrix} \alpha & 2 & 2 \\ -3 & 0 & 4 \\ 1 & -1 & 1 \end{bmatrix}$ is not invertible, then:
 (a) $\alpha = -5$ (b) $\alpha = 5$
 (c) $\alpha = 0$ (d) $\alpha = 1$ [NDA (II) - 2012]
17. If the matrix AB is a zero matrix, then which one of the following is correct?

- (a) A must be equal to zero matrix or B must be equal to zero matrix.
 (b) A must be equal to zero matrix and B must be equal to zero matrix.
 (c) It is not necessary that either A is zero matrixes or B is zero matrixes
 (d) None of the above

[NDA (II) - 2012]

18. If the sum the matrices $\begin{bmatrix} x \\ x \\ y \end{bmatrix}$, $\begin{bmatrix} y \\ y \\ z \end{bmatrix}$ and $\begin{bmatrix} z \\ 0 \\ 0 \end{bmatrix}$ is the matrix $\begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix}$,

then what is the value of y?

- (a) -5 (b) 0
 (c) 5 (d) 10

[NDA (II) - 2012]

19. A square matrix $[a_{ij}]$ such that $a_{ij} = 0$ for $i \neq j$ and $a_{ij} = k$ where k is a constant for $i = j$ is called.
 (a) Diagonal matrix but not scalar matrix
 (b) Scalar matrix
 (c) Unit matrix
 (d) None of the above

[NDA (II) - 2012]

20. If A and B are two non-singular square matrices such that $AB = A$, then which one of the following is correct?
 (a) B is an identity matrix
 (b) $B = A^{-1}$
 (c) $B = A^2$
 (d) Determinant of B is zero

[NDA (I) - 2013]

21. If $\begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$, then what is λ equal to ?
 (a) 7 (b) -7
 (c) 9 (d) -9

[NDA (II) - 2013]

22. If $A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ where $i = \sqrt{-1}$, then which one of the following is correct?
 (a) $AB = -C$
 (b) $AB = C$
 (c) $A^2 = B^2 = C^2 = I$, where I is the identity matrix
 (d) $BA \neq -C$

[NDA (II) - 2013]

23. Consider the following statements:
 I. The product of two non-zero matrices can never be identity matrix.
 II. The product of two non-zero matrices can never be zero matrix.
 Which of the above statements is/are correct?
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II

[NDA (II) - 2013]

24. Consider the following statements:

I. The matrix $\begin{pmatrix} 1 & 2 & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{pmatrix}$ is singular

II. The matrix $\begin{pmatrix} c & 2c & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{pmatrix}$, in non-singular

Which of the above statement is/are correct.

- (a) Only I (b) Only II
 (c) Both I and II (d) Either I or II

[NDA (II) - 2013]

25. If $2A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, then what is A^{-1} equal to?

- (a) $\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$ (b) $\frac{1}{2} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$
 (c) $\frac{1}{4} \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$ (d) None of these

[NDA (II) - 2013]

26. If A is a square matrix of order 3 with $|A| \neq 0$, then which one of the following is correct?

- (a) $|\text{adj } A| = |A|$ (b) $|\text{adj } A| = |A|^2$
 (c) $|\text{adj } A| = |A|^3$ (d) $|\text{adj } A|^2 = |A|$

[NDA(II)-2013]

27. Consider the following statements in respect of the matrix A

$$= \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix} :$$

I. The matrix A is skew symmetric

II. The matrix A is symmetric

III. The matrix A is invertible

Which of the above statements is /are correct?

- (a) Only I (b) Only III
 (c) I and II (d) II and III

[NDA (I) - 2014]

28. Consider two matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}$

which one of the following is correct?

- (a) B is the right inverse of A
 (b) B is the left inverse of A
 (c) B is the both sided inverse of A
 (d) None of the above

[NDA (I) - 2014]

29. If A is any matrix, then the product AA is defined only when A is a matrix of order $m \times n$, where:

- (a) $m > n$ (b) $m < n$
 (c) $m = n$ (d) $m \leq n$

[NDA (I) - 2014]

30. If, A and B be two matrices such that $AB = A$ and $BA = B$. Then, which of the following statements are correct?

1. $A^2 = A$
 2. $B^2 = B$
 3. $(AB)^2 = AB$
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) 1, 2 and 3

[NDA (II) - 2014]

31. If the matrix A is such that $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$, then what is A equal to?

- (a) $\begin{pmatrix} 1 & 4 \\ 0 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$
 (c) $\begin{pmatrix} -1 & 4 \\ 0 & -1 \end{pmatrix}$ (d) $\begin{pmatrix} 1 & -4 \\ 0 & -1 \end{pmatrix}$

[NDA (II) - 2014]

32. From the matrix equation $AB = AC$, where A, B and C are the square matrices of same order, we can conclude $B = C$ provided:

(a) A is non-singular
 (b) A is singular
 (c) A is symmetric
 (d) A is skew symmetric

[NDA (II) - 2014]

33. If $A = \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix}$ is symmetric, then what is x equal to?

(a) 2 (b) 3
 (c) -1 (d) 5

[NDA (II) - 2014]

34. The matrix $\begin{bmatrix} 0 & -4+i \\ 4+i & 0 \end{bmatrix}$ is:

(a) Symmetric (b) Skew-symmetric
 (c) Hermitian (d) Skew-hermitian

[NDA (I) - 2015]

35. Which one of the following matrices is an elementary matrix?

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 2 \end{bmatrix}$

[NDA (I) - 2015]

36. If $A = \begin{bmatrix} 2 & 7 \\ 1 & 5 \end{bmatrix}$, then what is $A + 3A^{-1}$ equal to?

(a) 3I (b) 5I
 (c) 7I (d) none of these

[NDA (I) - 2015]

37. If $X = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix}$ and $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ satisfy the equation $AX = B$, then the matrix A is equal to:

(a) $\begin{bmatrix} -7 & 26 \\ 1 & -5 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & 26 \\ 4 & 17 \end{bmatrix}$
 (c) $\begin{bmatrix} -7 & -4 \\ 26 & 13 \end{bmatrix}$ (d) $\begin{bmatrix} -7 & 26 \\ -6 & 23 \end{bmatrix}$

[NDA (I) - 2015]

38. Let $A = \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $C = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. If $AB = C$, then what is A^2 equal to?

(a) $\begin{bmatrix} 6 & -10 \\ 4 & 26 \end{bmatrix}$ (b) $\begin{bmatrix} -10 & 5 \\ 4 & 24 \end{bmatrix}$
 (c) $\begin{bmatrix} -5 & -6 \\ -4 & -20 \end{bmatrix}$ (d) $\begin{bmatrix} -5 & -7 \\ -5 & 20 \end{bmatrix}$

[NDA (I) - 2015]

39. If $E(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ then $E(\alpha)E(\beta)$ is equal to:

(a) $E(\alpha\beta)$ (b) $E(\alpha-\beta)$
 (c) $E(\alpha+\beta)$ (d) $-E(\alpha+\beta)$

[NDA (I) - 2015]

40. If $A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 4 \end{bmatrix}$, then the matrix X for which $2x + 3A = 0$ holds true is:

(a) $\begin{bmatrix} -\frac{3}{2} & 0 & -3 \\ -3 & -\frac{9}{2} & -6 \end{bmatrix}$ (b) $\begin{bmatrix} -\frac{3}{2} & 0 & -3 \\ 3 & -\frac{9}{2} & -6 \end{bmatrix}$
 (c) $\begin{bmatrix} \frac{3}{2} & 0 & 3 \\ 3 & -\frac{9}{2} & 6 \end{bmatrix}$ (d) $\begin{bmatrix} -\frac{3}{2} & 0 & 3 \\ -3 & \frac{9}{2} & -6 \end{bmatrix}$

[NDA (II) - 2015]

41. The matrix $A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{bmatrix}$ will have inverse for every real number x except for:

(a) $x = \frac{11 \pm \sqrt{5}}{2}$ (b) $x = \frac{9 \pm \sqrt{5}}{2}$
 (c) $x = \frac{11 \pm \sqrt{3}}{2}$ (d) $x = \frac{9 \pm \sqrt{3}}{2}$

[NDA (II) - 2015]

42. If $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$ then the following is/are correct?

1. A and B commute
 2. AB is a null matrix

Select the correct answer using the codes given below:

(a) Only 1 (b) Only 2
 (c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2015]

43. Consider the following in respect of the matrix $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$:

1. $A^2 = -A$
 2. $A^3 = 4A$

Which of the above is /are correct?

(a) Only 1 (b) Only 2
 (c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2016]

44. If A is a square matrix, then what is $\text{adj}(A^{-1}) - (\text{adj}A)^{-1}$ equal to?

(a) $2|A|$ (b) Null matrix
 (c) Unit matrix (d) None of these

[NDA (I) - 2016]

45. What is $[x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ equal to?

(a) $[ax + hy + gz \ h + b + f \ g + f + c]$
 (b) $\begin{bmatrix} a & h & g \\ hx & by & fz \\ g & f & c \end{bmatrix}$

$$(c) \begin{bmatrix} ax & hy & gz \\ hx & by & fz \\ gx & fy & cz \end{bmatrix}$$

$$(d) [ax + hy + gz \quad hx + by + fz \quad gx + fy + cz]$$

[NDA (II) - 2016]

46. If $m = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $n = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, then what is the value of the determinant of $m \cos \theta - n \sin \theta$?
- (a) -1 (b) 0
(c) 1 (d) 2

[NDA (II) - 2016]

47. If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then which of the following are correct?

- $f(\theta) \times f(\phi) = f(\theta + \phi)$.
 - The value of the determinant of the matrix $f(\theta) \times f(\phi)$ is 1.
 - The determinant of $f(x)$ is an even number.
- Select the correct answer using the codes given below:
- (a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2, and 3

[NDA (II) - 2016]

48. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$, then which of the following is/are correct?
- $AB (A^{-1} B^{-1})$ is a unit matrix.
 - $(AB)^{-1} = A^{-1} B^{-1}$
- Select the correct answer using the codes given below:
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2016]

49. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then what is AA^T equal to (where A^T is the transpose of A)?
- (a) Null matrix (b) Identity matrix
(c) A (d) $-A$

[NDA (I) - 2017]

50. $A = \begin{bmatrix} x+y & y \\ x & x-y \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ and $C = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$. If $AB = C$, then what is A^2 equal to?
- (a) $\begin{bmatrix} 4 & 8 \\ -4 & -16 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & -4 \\ 8 & -16 \end{bmatrix}$
(c) $\begin{bmatrix} -4 & -8 \\ 4 & 12 \end{bmatrix}$ (d) $\begin{bmatrix} -4 & -8 \\ 8 & 12 \end{bmatrix}$

[NDA (I) - 2017]

51. If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then what is A^3 equal to?
- (a) $\begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos^3 \theta & \sin^3 \theta \\ -\sin^3 \theta & \cos^3 \theta \end{bmatrix}$
(c) $\begin{bmatrix} \cos 3\theta & -\sin 3\theta \\ -\sin 3\theta & \cos \theta \end{bmatrix}$ (d) $\begin{bmatrix} \cos^3 \theta & \sin^3 \theta \\ -\sin^3 \theta & \cos^3 \theta \end{bmatrix}$

[NDA (I) - 2017]

52. What is the order of :

$$[x \ y \ z] \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} ?$$

- (a) 3×1 (b) 1×1
(c) 1×3 (d) 3×3

[NDA (I) - 2017]

53. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the value of A^4 is:

- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$
(c) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

[NDA (I) - 2017]

54. If $A = \begin{bmatrix} 4i-6 & 10i \\ 14i & 6+4i \end{bmatrix}$ and $k = \frac{1}{2i}$, where $i = \sqrt{-1}$ then kA is equal to:

- (a) $\begin{bmatrix} 2+3i & 5 \\ 7 & 2-3i \end{bmatrix}$ (b) $\begin{bmatrix} 2-3i & 5 \\ 7 & 5+3i \end{bmatrix}$
(c) $\begin{bmatrix} 2-3i & 7 \\ 5 & 3+3i \end{bmatrix}$ (d) $\begin{bmatrix} 2+3i & 5 \\ 7 & 2+3i \end{bmatrix}$

[NDA (II) - 2017]

55. If A is a square matrix, then the value of $\text{adj } A^T - (\text{adj } A)^T$ is equal to:
- (a) A
(b) $2|A|I$, where I is the identity matrix
(c) Null matrix whose order is same as that of A
(d) Unit matrix whose order is same as that of A

[NDA (II) - 2017]

56. If a, b, c are non-zero real numbers, then the inverse of the matrix, $A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$ is equal to:

- (a) $\begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$ (b) $\frac{1}{abc} \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$
(c) $\frac{1}{abc} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (d) $\frac{1}{abc} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$

[NDA (II) - 2017]

57. The adjoint of the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$ is:

- (a) $\begin{bmatrix} -1 & 6 & 2 \\ -2 & 1 & -4 \\ 6 & 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 6 & 1 & 2 \\ 4 & -1 & 2 \\ 6 & 3 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} -6 & 2 & 1 \\ 4 & -2 & 1 \\ 3 & 1 & -6 \end{bmatrix}$

[NDA (II) - 2017]

58. If $A = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}$, then which one of the following is correct?

- (a) $A^2 = -2A$ (b) $A^2 = -4A$
 (c) $A^2 = -3A$ (d) $A^2 = 4A$

[NDA (II) - 2017]

59. The matrix A has x rows and x + 5 columns. The matrix B has y rows and 11-y columns both AB and BA exist. What are the values of x and y respectively?

- (a) 8 and 3 (b) 3 and 4
 (c) 3 and 8 (d) 8 and 8

[NDA (II) - 2017]

60. If α and β are the roots of the equation $1 + x + x^2 = 0$, then the matrix product $\begin{bmatrix} 1 & \beta \\ \alpha & \alpha \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ 1 & \beta \end{bmatrix}$ is equal to:

- (a) $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}$
 (c) $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & -1 \\ -1 & -2 \end{bmatrix}$

[NDA (II) - 2017]

61. If $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ and $A^2 - kA - I_2 = O$, where I_2 is the 2×2 identity matrix, then what is the value of k?

- (a) 4 (b) -4
 (c) 8 (d) -8

[NDA (I) - 2018]

62. If A is a 2×3 matrix and AB is a 2×5 matrix, then B must be a

- (a) 3×5 matrix (b) 3×3 matrix
 (c) 3×2 matrix (d) 5×2 matrix

[NDA (I) - 2018]

63. What is the inverse of the matrix

$$A = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- (a) $\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}$
 (c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$ (d) $\begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

[NDA (I) - 2018]

64. Consider the following in respect of matrices A and B of same order:

I. $A^2 - B^2 = (A+B)(A-B)$

II. $(A-I)(I+A) = 0 \Leftrightarrow A^2 = I$

Where I is the identity matrix and O is the null matrix.

Which of the above is/are correct?

- (a) I only (b) 2 only
 (c) Both I and II (d) Neither 1 nor II

[NDA (II) - 2018]

65. Let matrix B be the adjoint of a square matrix A, ℓ be the identity matrix of same order as A. If k ($\neq 0$) is the determinant of the matrix A, then what is AB equal to?

- (a) ℓ (b) $k\ell$
 (c) $k^2\ell$ (d) $(1/k)\ell$

[NDA (II) - 2018]

66. Consider the following in respect of matrices A, B and C of same order:

I. $(A+B+C)' = A' + B' + C'$

II. $(AB)' = A'B'$

III. $(ABC)' = C'B'A'$

Where A' is the transpose of the matrix A. Which of the above are correct?

- (a) I and II only (b) II and III only
 (c) I and III only (d) I, II and III only

[NDA (II) - 2018]

67. What should be the value of x so that the matrix $\begin{pmatrix} 2 & 4 \\ -8 & x \end{pmatrix}$

does not have an inverse?

- (a) 16 (b) -16
 (c) 8 (d) -8

[NDA (II) - 2018]

68. If A and B are two invertible square matrices of same order, then what is $(AB)^{-1}$ equal to?

- (a) $B^{-1}A^{-1}$ (b) $A^{-1}B^{-1}$
 (c) $B^{-1}A$ (d) $A^{-1}B$

[NDA (II) - 2018]

69. What is the ad joint of the matrix:

$$\begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{pmatrix}$$

- (a) $\begin{pmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ (b) $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
 (c) $\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ (d) $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

[NDA (II) - 2018]

70. A square matrix A is called orthogonal if.

- (a) $A=A^2$ (b) $A'=A^{-1}$
 (c) $A=A^{-1}$ (d) $A=A'$

[NDA (II) - 2018]

71. If A is an identity matrix of order 3, then its inverse (A^{-1}) .

- (a) is equal to null matrix
 (b) is equal to A
 (c) is equal to 3A
 (d) does not exist

[NDA-2019(1)]

72. If $B = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 4 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ then what is adjoint of B equal to?

- (a) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & -1 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 8 \end{bmatrix}$
 (c) $\begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (d) Does not exist

[NDA (I) - 2019]

73. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ then the matrix A is

- (a) Singular matrix (b) Involutory matrix
 (c) Nilpotent matrix (d) Idempotent matrix

[NDA (I) - 2019]

74. If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ then the expression $A^3 - 2A^2$ is

- (a) a null matrix (b) an identity matrix
(c) equal to A (d) equal to $-A$

[NDA (II) - 2019]

75. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, then

Which one of the following is correct?

- (a) Both AB and BA exists
(b) Neither AB nor BA exists
(c) AB exists but BA not exists
(d) AB does not exists but BA exists

[NDA (II) - 2019]

76. Let $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$, $B = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ then what is AB equal to?

(a) $\begin{bmatrix} ax + hy + gz \\ y \\ z \end{bmatrix}$

(b) $\begin{bmatrix} ax + hy + gz \\ hx + by + fz \\ z \end{bmatrix}$

(c) $\begin{bmatrix} ax + hy + gz \\ hx + by + fz \\ gx + fy + cz \end{bmatrix}$

(d) $[ax + hy + gz \quad hx + by + fz \quad gx + fy + cz]$

[NDA 2020]

77. If A is a matrix of order 3×5 and B is a matrix of order 5×3 , then the order of AB and BA will respectively be

- (a) 3×3 and 3×3 (b) 3×5 and 5×3
(c) 3×3 and 5×5 (d) 5×3 and 3×5

[NDA 2020]

78. If matrix $A = \begin{bmatrix} 1-i & i \\ -i & 1-i \end{bmatrix}$ then which one the following is correct?

- (a) A is hermition
(b) A is skew hermition
(c) $(\overline{A})^T + A$ is hermition
(d) $(\overline{A})^T + A$ is skew hermition

[NDA 2020]

79. For how many values of k, is the matrix $\begin{bmatrix} 0 & k & 4 \\ -k & 0 & -5 \\ -k & k & -1 \end{bmatrix}$ is

singular?

- (a) Only one (b) Only two
(c) Only 4 (d) Infinite

[NDA 2020]

80. Let $A = \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $C = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ If $AB=C$, then

what is the value of determinant of A?

- (a) -10 (b) -14
(c) -24 (d) -34

[NDA 2020]

81. Consider the following in respect of a non singular matrix of order 3

I. $A(\text{adj}A) = (\text{adj}A)A$

II. $|\text{adj} A| = |A|$

Where I is the identity matrix and O is the null matrix.

Which of the above is/are correct?

- (a) I only (b) 2 only
(c) Both I and II (d) Neither 1 nor II

[NDA 2020]

82. If A and B are two matrices such that AB of order $n \times n$, then which one of the following is correct?

- (a) A and B should be square matrices of same order
(b) Either A or B should be a square matrix.
(c) Both A and B should be same order
(d) Order of A and B need not be same.

[NDA (I) 2021]

83. How many matrices of different orders are possible with elements comprising all prime numbers less than 30?

- (a) 2 (b) 3
(c) 4 (d) 6

[NDA (I) 2021]

84. If $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ where $A \in \mathbb{N}$, then what is $A^{100} - A^{50} - 2A^{25}$ equal to?

- (a) $-2I$ (b) $-I$
(c) $2I$ (d) I

Where I is the identify matrix.

[NDA (II) 2021]

85. How many distinct matrices exist with all four entries taken from $\{1,2\}$?

- (a) 16 (b) 24
(c) 32 (d) 48

[NDA (II) 2021]

86. Consider the following in respect of the matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

1. Inverse of A does not exist

2. $A^3 = A$

3. $3A = A^2$

Which of the above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (II) 2021]

87. The inverse of a matrix A is given by $\begin{bmatrix} -2 & 1 \\ 3 & -\frac{1}{2} \end{bmatrix}$. What is

A equal to?

- (a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$

[NDA (II) 2021]

88. Let $A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$ and $(mI + nA)^2 = A$ where m, n are positive real numbers and I is the identity matrix. What is $(m + n)$ equal to:

- (a) 0 (b) $\frac{1}{2}$

- (c) 1 (d) $\frac{5}{2}$

[NDA (II) 2021]

89. Consider the following in respect of the matrices:

$$A = \begin{bmatrix} m & n \\ -n & -m \end{bmatrix}, B = \begin{bmatrix} -n & -m \\ m & n \end{bmatrix} \text{ and } C = \begin{bmatrix} m \\ -m \end{bmatrix}$$

1. $CA = CB$
2. $AC = BC$
3. $C(A + B) = CA + CB$

Which of the above statement is/are correct?

- (a) 1 only (b) 2 only
(c) 2 and 3 (d) 1 and 2

[NDA (I) - 2022]

90. If $A = \begin{bmatrix} 2 \sin \theta & \cos \theta & 0 \\ -2 \cos \theta & \sin \theta & 0 \\ -1 & 1 & 1 \end{bmatrix}$, then what is $A (\text{adj. } A)$ equal to?

- (a) Null matrix (b) $-I$
(c) I (d) $2I$

Where I is the identity matrix.

[NDA (I) - 2022]

91. For what value of k is the matrix

$$\begin{bmatrix} 2 \cos 2\theta & 2 \cos 2\theta & 6 \\ 1 - 2 \sin^2 \theta & 2 \cos^2 \theta - 1 & 3 \\ k & 2k & 1 \end{bmatrix} \text{ singular?}$$

- (a) 0 only (b) 1 only
(c) 2 only (d) Any real value

[NDA (I) - 2022]

92. Let A be a non-singular matrix and $B = \text{adj} A$. Which of the following statement is/are correct?

1. $AB = BA$
2. AB is a scalar matrix
3. AB can be null matrix

Select the correct answer using the code given below:

- (a) 1 only (b) 1 and 2 only
(c) 2 only (d) 1, 2 and 3

[NDA (I) - 2022]

93. Consider the following statements in respect of square matrices A and B of same order

1. If AB is a null matrix, then at least one of A and B is a null matrix

2. If AB is an identity matrix, then $BA = AB$

Which of the above statement is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) neither 1 nor 2

[NDA (I) - 2022]

94. If A is the identity matrix of order 3 and B is its transpose, then what is the value of determinant of the matrix $C = A + B$?

- (a) 1 (b) 2
(c) 4 (d) 8

[NDA (I) - 2022]

95. Let A and B be non-singular matrices of the same order such that $AB = A$ and $BA = B$. Which of the following statements is/are correct?

1. $A^2 = A$
2. $AB^2 = A^2B$

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

Consider the following for the next three (03) items that follow:

$$\text{Let } A = \begin{pmatrix} 0 & \sin^2 \theta & \cos^2 \theta \\ \cos^2 \theta & 0 & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta & 0 \end{pmatrix} \text{ and } A = P + Q \text{ where } P \text{ is}$$

symmetric matrix and Q is skew-symmetric matrix.

96. What is P equal to?

$$(a) \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

$$(b) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$(c) \cos 2\theta \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$(d) \cos 2\theta \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & 1/2 & 0 \end{pmatrix}$$

[NDA 2022 (II)]

97. What is Q equal to?

$$(a) \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

$$(b) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$(c) \cos 2\theta \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$(d) \cos 2\theta \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & -1/2 \\ -1/2 & 1/2 & 0 \end{pmatrix}$$

[NDA 2022 (II)]

98. What is the minimum value of determinant of A ?

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) 1

[NDA 2022 (II)]

99. If $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & \sin \theta & -\cos \theta \end{pmatrix}$, then which of the following are correct?

1. $A + \text{adj} A$ is a null matrix
2. $A^{-1} + \text{adj} A$ is null matrix
3. $A - A^{-1}$ is a null matrix

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA 2022 (II)]

100. If X is a matrix of order 3×3 , Y is a matrix of order 2×3 and Z is a matrix of order 3×2 , then which of the following are correct?

1. $(ZY)X$ is a square matrix having 9 entries
2. $Y(XZ)$ is a square matrix having 4 entries
3. $X(YZ)$ is not defined

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA 2022 (II)]

101. Let A and B be symmetric matrices of same order, then which one of the following is correct regarding $(AB - BA)$?

1. Its diagonal entries are equal but nonzero

2. The sum of its non-diagonal entries is zero
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA – 2023 (1)]

- 102.** Consider the following statements in respect of square matrices A, B, C each of same order n:
1. $AB = AC \Rightarrow B = C$ if A is non singular
2. If $BX = CX$ for every column matrix X having n rows then $B = C$.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA – 2023 (1)]

- 103.** Let $AX = B$ be a system of 3 linear equations with 3-unknowns. Let X_1 and X_2 be its two distinct solutions. If the combination $aX_1 + bX_2$ is a solution of $AX = B$; where a, b are real numbers, then which one of the following is correct?
(a) $a = b$ (b) $a + b = 1$
(c) $a + b = 0$ (d) $a - b = 1$

[NDA – 2023 (1)]

- 104.** If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then what is the value of $\det(I + AA')$, where I is the 3×3 identity matrix?
(a) 15 (b) 6
(c) 0 (d) -1

[NDA-2023 (2)]

- 105.** If $A = \begin{bmatrix} 0 & 3 & 4 \\ -3 & 0 & 5 \\ -4 & -5 & 0 \end{bmatrix}$, then which one of the following statements is correct?
(a) A^2 is symmetric matrix with $\det(A^2) = 0$
(b) A^2 is symmetric matrix with $\det(A^2) \neq 0$.
(c) A^2 is skew-symmetric matrix with $\det(A^2) = 0$
(d) A^2 is skew-symmetric matrix with $\det(A^2) \neq 0$

[NDA-2023 (2)]

- 106.** If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, then which one of the following statements are correct?
1. A^n will always be singular for any positive integer n
2. A^n will always be a diagonal matrix for any positive integer n
3. A^n will always be a symmetric matrix for any positive integer n.
Select the correct answer using the code given below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2023 (2)]

- 107.** If $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, then what is the value of $\det[\text{adj}(\text{adj}A)]$?
(a) 5 (b) 25
(c) 125 (d) 625

[NDA-2023 (2)]

- 108.** If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then what is $23A^2 - 19A^2 - 4A$ equal to?

- (a) null matrix of order 3 (b) identity matrix of order 3
(c) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$

[NDA-2023 (2)]

- 109.** What is the number of different matrices, each having 4 entries that can be formed using 1, 2, 4, 5, (repetition is allowed)?
(a) 72 (b) 216
(c) 254 (d) 768

[NDA-2024 (1)]

- 110.** If $A = \begin{bmatrix} \sin 2\theta & 2\sin^2 \theta - 1 & 0 \\ \cos 2\theta & 2\sin \theta \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then which of the following statements is/are correct?
1. $A^{-1} = \text{adj}A$
2. A is skew-symmetric matrix
3. $A^{-1} = A^T$
Select the correct answer using the code given below:
(a) 1 only (b) 1 and 2
(c) 1 and 3 (d) 2 and 3

[NDA-2024 (1)]

- 111.** Consider the following statements in respect of two non-singular matrices A and B are of same order n :
1. $\text{adj}(AB) = (\text{adj}A)(\text{adj}B)$
2. $\text{adj}(AB) = \text{adj}(BA)$
3. $(AB)\text{adj}(AB) - |AB|I_n$ is a null matrix of order n
How many of the above statements are correct?
(a) None (b) Only one statement
(c) Only two statements (d) All three statements

[NDA-2024 (1)]

- 112.** Consider the following statements in respect of a non-singular matrix A of order n :
1. $A(\text{adj}A^T) = A(\text{adj}A)^T$
2. If $A^2 = A$, then A is identity matrix of order n
3. If $A^3 = A$, then A is identity matrix of order n
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2024 (1)]

- 113.** Consider the following statements in respect of a skew-symmetric matrix A of order 3:
1. All diagonal elements are zero
2. The sum of all the diagonal elements of the matrix is zero
3. A is orthogonal matrix
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2024 (1)]

Direction: Consider the following for the two items given below

Let $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

- 114.** What is $A(\text{adj}A)$ equal to ?

- (a) $\begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$

(d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

[NDA-2024 (2)]

 115. What is A^{-1} equal to?

(a) $\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$

(b) $\begin{bmatrix} 1/2 & -1/2 & 0 \\ -1 & 3/2 & -2 \\ -1 & 3/2 & -3/2 \end{bmatrix}$

(c) $\begin{bmatrix} 2 & -2 & 0 \\ -4 & 6 & -8 \\ -4 & 6 & -6 \end{bmatrix}$

(d) $\begin{bmatrix} 1/5 & -1/5 & 0 \\ -2/5 & 3/5 & -4/5 \\ -2/5 & 3/5 & -3/5 \end{bmatrix}$

[NDA-2024 (2)]

116. Consider the following in respect of matrices

$$P = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \text{ and } Q = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

I. PQ is null matrix.

II. QP is an identity matrix of order 3

III. PQ = QP

Which of the above is/are correct?

- (a) I only (b) II only
 (c) I and III (d) II and III

[NDA-2024 (2)]

 117. Let X be a matrix of order 3×3 , Y be a matrix of order 2×3 and Z be a matrix of order 3×2 . Which of the following statements are correct?

- I. (ZY)X is defined and is a square matrix of order 3
 II. Y(XZ) is defined and is a square matrix of order 2
 III. X(YZ) is not defined.

Select the answer using the code given below

- (a) I and II only (b) II and III only
 (c) I and III only (d) I, II and III

[NDA-2024 (2)]

118. Let A and B be two square matrices of same order. If AB is a null matrix, then which one of the following is correct?

- (a) Both A and B are null matrices
 (b) Either A or B is a null matrix
 (c) B is a null matrix if A is a non-singular matrix
 (d) Both A and B are singular matrices

[NDA-2024 (2)]

119. If

$$\begin{bmatrix} x & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = [45]$$

Then which one of the following is a value of x?

- (a) -2 (b) -1
 (c) 0 (d) 1

[NDA-2025 (1)]

120. If

$$A = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix}$$

 where x, y, z are integers, is an orthogonal matrix, then what is the value of $x^2 + y^2 + z^2$?

- (a) 0 (b) 1
 (c) 2 (d) 3

[NDA-2025 (1)]

121. Consider the following in respect of a non-singular matrix M:

 I. $|M^2| = |M|^2$

 II. $|M| = |M^{-1}|$

 III. $|M| = |M^T|$

How many of the above are correct?

- (a) None (b) One
 (c) Two (d) All three

[NDA-2025 (1)]

 122. If $f(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

 Then, what is $\{f(\pi)\}^2$ equal to?

- (a) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
 (c) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[NDA-2025 (1)]

 123. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then what is $A^2 - 4A$ is equal to?

- (a) $-5I_3$ (b) $-I_3$
 (c) I_3 (d) $5I_3$

 Where, I_3 is the identity matrix of order 3

[NDA-2025 (1)]

 124. If $A = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix}$

 Where x, y, z are integers, is an orthogonal matrix, then what is A^2 equal to?

- (a) Null matrix (b) Identity matrix
 (c) A (d) -A

[NDA-2025 (1)]

125. Consider the following in respect of non-singular matrices A and B:

 I. $(AB)^{-1} = A^{-1} B^{-1}$

 II. $(BA)(AB)^{-1} = I$, where I is the identity matrix

 III. $(AB)^T = A^T B^T$

How many of the above are correct?

- (a) None (b) One
 (c) Two (d) All three

[NDA-2025 (2)]

ANSWER KEY

1.	a	2.	b	3.	d	4.	c	5.	a	6.	d	7.	b	8.	b	9.	c	10.	b
11.	a	12.	b	13.	a	14.	c	15.	c	16.	a	17.	c	18.	b	19.	b	20.	a
21.	b	22.	a	23.	d	24.	a	25.	d	26.	b	27.	a	28.	b	29.	c	30.	d
31.	a	32.	a	33.	d	34.	d	35.	b	36.	c	37.	a	38.	a	39.	c	40.	d
41.	a	42.	b	43.	b	44.	b	45.	d	46.	c	47.	a	48.	d	49.	b	50.	d
51.	a	52.	b	53.	a	54.	a	55.	c	56.	a	57.	b	58.	b	59.	c	60.	b
61.	a	62.	a	63.	a	64.	b	65.	b	66.	c	67.	b	68.	a	69.	a	70.	b
71.	b	72.	a	73.	b	74.	a	75.	c	76.	c	77.	c	78.	c	79.	d	80.	b
81.	a	82.	d	83.	c	84.	a	85.	d	86.	c	87.	a	88.	d	89.	c	90.	d
91.	d	92.	b	93.	b	94.	d	95.	c	96.	a	97.	d	98.	a	99.	d	100.	d
101.	b	102.	c	103.	b	104.	a	105.	a	106.	b	107.	d	108.	a	109.	d	110.	c
111.	b	112.	a	113.	a	114.	d	115.	a	116.	c	117.	d	118.	c	119.	d	120.	b
121.	c	122.	d	123.	d	124.	b	125.	a										

Solutions

Sol.1. (a)

For the singular matrix,

$$\begin{vmatrix} 2-x & 1 & 1 \\ 1 & 3-x & 0 \\ -1 & -3 & -x \end{vmatrix} = 0 \text{ (expand with respect to } R_1)$$

$$\Rightarrow (2-x)[x(x-3)] - [-x] + [-3 + (3-x)] = 0$$

$$\Rightarrow x(x-3)(x-2) = 0$$

$$\Rightarrow x = 0, 2, 3$$

So, the solution set is, $S = \{0, 2, 3\}$

Sol.2. (b)

$$A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix}$$

Given, $A^2 = B$

$$\begin{bmatrix} \alpha^2 & 0 \\ \alpha+1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

On comparing, $\alpha + 1 = 2$

$$\Rightarrow \alpha = 1$$

Sol.3. (d)

Given matrices are,

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}_{2 \times 2}, B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}_{2 \times 2}$$

Since, the order of both matrices are same

So, AB and BA is well defined

Now, $AB =$

$$\begin{bmatrix} 3 \times 1 + 1 \times 0 & 3 \times 1 + 1 \times 2 \\ 0 \times 1 + 4 \times 0 & 0 \times 1 + 4 \times 2 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 0 & 8 \end{bmatrix}$$

$$\text{and } AB = \begin{bmatrix} 1 \times 3 + 1 \times 0 & 1 \times 1 + 4 \times 1 \\ 3 \times 0 + 2 \times 0 & 0 \times 1 + 2 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 0 & 8 \end{bmatrix}$$

$$\Rightarrow AB = BA$$

Sol.4. (c)

I. If the inverse of a square matrix exists, then it will be unique

II. If two matrices A and B are singular of order n , then the product of both matrix is also a singular matrix of same order.

Sol.5. (a)

$$A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix} = I$$

$$\Rightarrow \begin{bmatrix} \alpha^2 + \beta^2 & 2\alpha\beta \\ 2\alpha\beta & \alpha^2 + \beta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \alpha^2 + \beta^2 = 1, 2\alpha\beta = 0$$

$$\Rightarrow \alpha = 0, \beta = 1 \text{ or } \alpha = 1, \beta = 0$$

Sol.6. (d)

$$\text{Given, } (1, 2, 3)_{1 \times 3} \begin{pmatrix} 1 & 3 & 0 \\ 3 & 0 & 2 \\ 2 & 0 & 1 \end{pmatrix}_{3 \times 3} \begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix}_{3 \times 1} = (0)_{1 \times 1}$$

$$\Rightarrow (1+9+4 \quad 3+0+0 \quad 0+6+2)_{1 \times 3} \begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix}_{3 \times 1} = (0)_{1 \times 1}$$

$$\Rightarrow (14 \ 3 \ 8)_{1 \times 3} \begin{pmatrix} 0 \\ 3 \\ x \end{pmatrix}_{3 \times 1} = (0)_{1 \times 1}$$

$$\Rightarrow (0+9+8x) = (0)$$

$$\Rightarrow (8x+9) = (0)$$

On comparing,

$$8x+9=0 \Rightarrow x = -\frac{9}{8}$$

Sol.7. (b)

I. Every zero-matrix is not necessarily square matrix

II. A matrix does not have a numerical value while every determinant have a numerical value

III. Unit matrix is a diagonal matrix and scalar matrix also.

Only statement III is correct.

Sol.8. (b)

We know that, every matrix possesses a unique inverse. Hence, B and C should be equal.

Sol.9. (c)

Given condition is $A^2 = I$

$$\Rightarrow A^{-1} \cdot A^2 = A^{-1} \cdot I$$

$$\Rightarrow A^{-1} (A \cdot A) = A^{-1} \cdot I \quad (\because A \cdot I = A)$$

$$\Rightarrow (A^{-1} A) \cdot A = A^{-1} \cdot I$$

$$\Rightarrow I \cdot A = A^{-1} \quad (\because A^{-1} A = I)$$

$$\Rightarrow A^{-1} = A$$

Sol.10. (b)

Here,

$$[x \ y \ z]_{1 \times 3} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}_{3 \times 3} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1}$$

$$\text{Order of matrix} = 1 \times 3 : 3 \times 3 : 3 \times 1 = 1 \times 1$$

Sol.11. (a)

The sum and product of matrices A and B exist, if A and B are square matrices of same order

It is not necessary that A and B are non-singular matrices for addition and product of two matrices.

Sol.12. (b)

$$\text{Given, } A = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$$

$$\text{Here, } |A| = \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1$$

$$\text{adj. } (A) = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj. } A}{|A|} = -1 \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\text{Here, } |B| = \begin{vmatrix} 0 & -1 \\ 1 & -2 \end{vmatrix} = 0 - (-1) = 1$$

$$\text{adj. } (B) = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$$

$$B^{-1} = \frac{\text{adj. } B}{|B|} = 1 \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}; B^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\therefore B^{-1} A^{-1} =$$

$$\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1+1 & 4-1 \\ 1+0 & -2+0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 3 \\ 1 & -2 \end{bmatrix}$$

Sol.13. (a)

$$AB = A$$

Multiply with B at left side

$$BAB = BA$$

$$(BA)B = BA$$

$$B^2 = B$$

Sol.14. (c)

B is the transpose of A

Sol.15. (c)

We know that by the property of diagonal matrix

At $A = \text{diagonal } (a_1, a_2, a_3)$

Then, $A^{-1} = \text{Inverse of } A$

$$= \text{Diagonal } (a_1^{-1}, a_2^{-1}, a_3^{-1})$$

$$= \text{Diagonal } \left(\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3} \right)$$

Hence, the inverse of diagonal matrix is a diagonal matrix.

Sol.16. (a)

If the matrix is not invertible, then it is necessary that the value of the determinant of this matrix must be zero.

$$\text{i.e., } \begin{vmatrix} \alpha & 2 & 2 \\ -3 & 0 & 4 \\ 1 & -1 & 1 \end{vmatrix} = 0$$

$$\begin{pmatrix} C_2 \rightarrow C_2 + C_1 \\ C_3 \rightarrow C_3 - C_1 \end{pmatrix}$$

$$\begin{vmatrix} \alpha & 2+\alpha & 2-\alpha \\ -3 & -3 & 7 \\ 1 & 0 & 0 \end{vmatrix} = 0$$

Expanding the determinant along the third row we get

$$1 \begin{vmatrix} 2+\alpha & 2-\alpha \\ -3 & 7 \end{vmatrix} = 0$$

$$\Rightarrow 14 + 7\alpha + 3(2-\alpha) = 14 + 7\alpha + 6 - 3\alpha = 0$$

$$\Rightarrow 20 + 4\alpha = 0$$

$$\Rightarrow 4\alpha = -20 \Rightarrow \alpha = -5$$

Sol.17. (c)

For the matrix AB is a zero matrix. It is not necessary that either A is zero matrix or B is zero matrix.

$$\text{e.g., Let } A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\therefore AB = 0, \text{ where } A, B \neq 0$$

Sol.18. (b)

$$\begin{bmatrix} x \\ x \\ y \end{bmatrix} + \begin{bmatrix} y \\ y \\ z \end{bmatrix} + \begin{bmatrix} z \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x & +y & +z \\ x & +y & +0 \\ y & +z & +0 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 5 \end{bmatrix}$$

$$\Rightarrow x + y + z = 10 \quad \dots(i)$$

$$x + y = 5 \quad \dots(ii)$$

$$y + z = 5 \quad \dots(iii)$$

From eqs. (i) and (iii),

$$x + (5) = 10 \Rightarrow x = 5$$

On putting the value of x in eq. (ii), we get

$$5 + y = 5 \Rightarrow y = 0$$

Sol.19. (b)

$$\text{Given, } [a_{ij}] = \begin{cases} a_{ij} = 0, \text{ for } i = j \\ a_{ij} = k, \text{ for } i \neq j \end{cases}$$

Where k is a constant

$$\therefore [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} \text{ let order } 3 \times 3$$

$$= \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}_{3 \times 3} = \text{scalar matrix}$$

Sol.20. (a)

Given that A and B are two non-singular square matrices. So, its inverse i.e., A^{-1} and B^{-1} must exist.

We have, $AB = A$

(A^{-1}) operating in left side on both sides, we get

$$A^{-1}(AB) = (A^{-1})A$$

$$\Rightarrow (A^{-1}A)B = (A^{-1}A) \quad (\because AA^{-1} = I \text{ and } B^{-1}B = I)$$

$$\Rightarrow IB = I \Rightarrow B = I = \text{identity matrix}$$

Sol.21. (b)

$$\text{Given, } \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 10 & -9-4+3 \\ 20 & -3-8+1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & -1 \\ 17 & -7 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 17 & \lambda \end{bmatrix}$$

On comparing, we get $\lambda = -7$.

Sol.22. (a)

Given that,

$$A = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}, B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ and } C = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$$

$$\text{Now, } AB = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = -C$$

Sol.23. (d)

We know that the product of two identity matrix are always an identity matrix, which is non-zero matrices.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 \\ 0+0 & 1+0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I = \text{Identity matrix}$$

The product of two non-zero matrices can sometimes be zero matrix.

$$\begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \times \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

$$= \begin{bmatrix} a+abc+bac & 0+b^2c-b^2c & 0+bc^2-bc^2 \\ a^2c+a+a^2c & -abc+0+abc & -ac^2+0+ac^2 \\ a^2b-a^2b+0 & ab^2-ab^2+0 & abc-abc+0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0 = \text{Zero matrix}$$

So, the statements are incorrect

Sol.24. (a)

$$(i) \text{ Let } A = \begin{bmatrix} 1 & 2 & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{bmatrix}$$

$$\text{Now, } |A| = 1(2a-2b) - 2(a-b) + 1(2ab-2ab) = 2a-2b+2b+0 = 0$$

i.e., A is a singular matrix

$$(ii) \text{ Let } B = \begin{bmatrix} c & 2c & 1 \\ a & 2a & 1 \\ b & 2b & 1 \end{bmatrix}$$

$$\text{Now, } |B| = c(2a-2b) - 2c(a-b) + 1(2b-2ab) = 2ac-2bc-2ac+2bc+0 = 0$$

Which is also represents a singular matrix

So, statement I is correct but statement II is incorrect.

Sol.25. (d)

$$\text{Given, that, } 2A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$$

$$\Rightarrow A = \frac{1}{2} \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 3/2 & 1 \end{bmatrix}$$

Now, $\text{adj } A =$

$$\begin{bmatrix} 1 & -3/2 \\ -1/2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ 3/2 & 1 \end{bmatrix}$$

$$\text{and } |A| = 1 - 3/4 = 1/4$$

$$\therefore A^{-1} =$$

$$\frac{\text{adj } A}{|A|} = 4 \begin{bmatrix} 1 & -1/2 \\ -3/2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -6 & 4 \end{bmatrix}$$

Sol.26. (b)

$$|\text{adj } A| = |A|^{(n-1)} = |A|^2$$

Sol.27. (a)

Given matrix,

$$A = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix}$$

Now, $A^T =$

$$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 3 \\ 2 & -3 & 0 \end{bmatrix}$$

$$= -\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -2 \\ -2 & 3 & 0 \end{bmatrix} = -A$$

$$\Rightarrow A = A^T$$

So, A is skew-symmetric matrix

$$\text{and } |A| = \begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ -2 & 3 & 0 \end{vmatrix} = 0-1(0-6)+2(-3)$$

$$= 6-6=0$$

Since, $|A| = 0$ i.e., A is singular matrix

So, A cannot be an invertible matrix.

Sol.28. (b)

Given matrices,

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}$$

$$\text{I. } AB = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}_{2 \times 3}$$

$$= \begin{bmatrix} 1+4 & 2+2 & -4-8 \\ 2+2 & 4+1 & -8-4 \\ 1+2 & 2+1 & -4-4 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 4 & -12 \\ 4 & 5 & -12 \\ 3 & 3 & 8 \end{bmatrix}_{3 \times 3}$$

$$\text{II. } BA = \begin{bmatrix} 1 & 2 & -4 \\ 2 & 1 & -4 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}_{3 \times 2}$$

$$= \begin{bmatrix} 1+4-4 & 2+2-4 \\ 2+2-4 & 4+1-4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

Now, we observe that B is not the right inverse of A but B is the left inverse of A.

Sol.29. (c)

Given that, A is any matrix.

Then, the product AA is defined only when A is a matrix of order $m \times n$ where, $m = n$ i.e., A must be a square matrix.

$$A \times A = (m \times n)(m \times n)$$

$$= (m \times n)(n \times n), \text{ if } m = n$$

$$= m \times n = n \times n \text{ or } m \times n$$

$\therefore$ A is a square matrix.

Sol.30. (d)

We have, $AB = A$

$$\therefore A^2 = (AB)(BA) = A(BA)B$$

$$= (AB)B \quad (\because BA=B)$$

$$= AB \quad (\because AB=A)$$

$$= A$$

$$\text{Also, } B^2 = (BA)(BA)$$

$$= B(AB)A \quad (\because AB=A)$$

$$= (B.A)A \quad (\because BA=B)$$

$$= B.A$$

$$= B$$

$$\text{Again, } (AB)^2 = (AB)(AB)$$

$$= A(BA)B$$

$$= (A.B).B \quad (\because BA=B)$$

$$= AB \quad (\because AB=A)$$

Sol.31. (a)

$$\therefore \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} A = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

$$\text{Let } B = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \text{ and } |B| = 1$$

$$\Rightarrow B^{-1} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} \left(\because \frac{1}{|B|} \text{adj } B \right)$$

$$\therefore A = B^{-1} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 0 & -1 \end{pmatrix}$$

Sol.32. (a)

From the matrix equation $AB = AC$, where A, B and C are the square matrices of same order.

We can conclude $B=C$ provided A is non singular.

Sol.33. (d)

$$\therefore A = A' \quad (A \text{ is symmetric})$$

$$\Rightarrow \begin{pmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{pmatrix} = \begin{pmatrix} 4 & 2x-3 \\ x+2 & x+1 \end{pmatrix}$$

$$\Rightarrow 2x-3 = x+2$$

$$\therefore x = 5$$

Sol.34. (d)

A square matrix A is said to be skew-hermitian,

$$\text{If } A^* = -A$$

given matrix is skew-hermitian matrix.

Sol.35. (b)

$$\begin{bmatrix} 1 & 5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is an elementary matrix. Since, the}$$

value of determinant of the given matrix is 1

Sol.36. (c)

$$\text{We have, } A = \begin{bmatrix} 2 & 7 \\ 1 & 5 \end{bmatrix}$$

$$\therefore |A| = 10 - 7 = 3$$

$$\text{Now, } A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$\therefore A^{-1} = \frac{1}{3} \begin{bmatrix} 5 & -1 \\ -7 & 2 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 5 & -7 \\ -1 & 2 \end{bmatrix}$$

$$\therefore A + 3A^{-1} = \begin{bmatrix} 2 & 7 \\ 1 & 5 \end{bmatrix} + 3 \times \frac{1}{3} \begin{bmatrix} 5 & -7 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = 7I$$

Sol.37. (a)

$$\begin{aligned} \therefore AX &= B \quad (\text{given}) \\ \therefore \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} &= \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 3p+q & -4p-q \\ 3r+s & -4r-s \end{bmatrix} &= \begin{bmatrix} 5 & 2 \\ -2 & 1 \end{bmatrix} \\ \Rightarrow 3p+q &= 5 \text{ and } -4p-q = 2 \\ \Rightarrow -p &= 7 \Rightarrow p = -7 \\ \text{Now, } q &= 5+21=26, 3r+s = -2 \\ \text{Also, } -4r-s &= 1 \\ \Rightarrow -r &= -1 \\ \Rightarrow r &= 1 \\ \text{and } S &= -2-3 = -5 \\ \therefore A &= \begin{bmatrix} -7 & 26 \\ 1 & -5 \end{bmatrix} \end{aligned}$$

Sol.38. (a)

$$\begin{aligned} \text{We have, } AB &= C \\ \therefore \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 2x+2y & -y \\ 4x & -x+y \end{bmatrix} &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 2x+y \\ 3x+y \end{bmatrix} &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \\ \Rightarrow 2x+y &= 3 \text{ and } 3x+y = 2 \\ \Rightarrow x &= 2-3 = -1 \\ \therefore y &= 5 \\ \text{Now, } A^2 &= \\ \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}^2 &= \begin{bmatrix} 4 & 5 \\ -2 & -6 \end{bmatrix} \begin{bmatrix} 4 & 5 \\ -2 & -6 \end{bmatrix} \\ &= \begin{bmatrix} 16-10 & 20-30 \\ -8+12 & -10+36 \end{bmatrix} = \begin{bmatrix} 6 & -10 \\ 4 & 26 \end{bmatrix} \end{aligned}$$

Sol.39. (c)

$$\begin{aligned} \text{Given, } E(\theta) &= \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \\ \therefore E(\alpha) &= \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \\ \text{and } E(\beta) &= \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix} \\ \text{Now, } E(\alpha) E(\beta) &= \\ \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & \sin \beta \\ -\sin \beta & \cos \beta \end{bmatrix} &= \\ \begin{bmatrix} \cos 2 \cdot \cos \beta - \sin 2 \cdot \sin \beta & \cos 2 \cdot \sin \beta + \sin 2 \cdot \cos \beta \\ \sin 2 \cdot \cos \beta - \sin \beta \cdot \sin 2 & -\sin 2 \cdot \sin \beta + \cos 2 \cdot \cos \beta \end{bmatrix} &= \\ \begin{bmatrix} \cos(\alpha+\beta) & \sin(\alpha+\beta) \\ -\sin(\alpha+\beta) & \cos(\alpha+\beta) \end{bmatrix} &= E(\alpha+\beta) \end{aligned}$$

Sol.40. (d)

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 4 \end{bmatrix} \\ \text{So, } 2x + 3A &= 0, x = ? \\ \Rightarrow 2x + 3 \begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 4 \end{bmatrix} &= 0 \\ \Rightarrow 2x + \begin{bmatrix} 3 & 0 & -6 \\ 6 & -9 & 12 \end{bmatrix} &= 0 \\ \Rightarrow 2x &= \begin{bmatrix} -3 & 0 & 6 \\ -6 & 9 & -12 \end{bmatrix} \end{aligned}$$

$$\Rightarrow x = \begin{bmatrix} -\frac{3}{2} & 0 & 3 \\ -3 & \frac{2}{9} & -6 \end{bmatrix}$$

Sol.41. (a)

$$\begin{aligned} A &= \begin{bmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{bmatrix} \\ \text{Take } |A| &= 0 \\ \Rightarrow \begin{vmatrix} 1 & 3 & 2 \\ 1 & x-1 & 1 \\ 2 & 7 & x-3 \end{vmatrix} &= 0 \end{aligned}$$

$$\begin{aligned} \text{After solving the matrix,} \\ \Rightarrow 1[(x-1)(x-3)-7] - 3(x-3) + 2(7-2x+2) &= 0 \\ \Rightarrow x^2 - 4x + 3 - 7 - 3x + 15 - 4x + 18 &= 0 \\ \Rightarrow x^2 - 4x - 4 - 3x - 4x + 33 &= 0 \\ \Rightarrow x^2 - 11x + 29 &= 0 \\ \therefore x &= \frac{11 \pm \sqrt{121-116}}{2} \\ &= \frac{11 \pm \sqrt{5}}{2} \end{aligned}$$

(from Sridhara Acharya formula)

$$\text{Or } x = \frac{11 \pm \sqrt{5}}{2}$$

Sol.42. (b)

$$\begin{aligned} A &= \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix}, B = \begin{bmatrix} -1 & -2 & -1 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix} \\ \Rightarrow AB &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

$$\Rightarrow BA \neq O$$

 So, $AB \neq BA$

 Thus only 2nd statement is correct.

Sol.43. (b)

$$\begin{aligned} A &= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \\ A.A &= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \\ A^2 &= \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \\ A^2 &= 2 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow A^2 = -2A \end{aligned}$$

 $\therefore$ Statement 1 is not correct

Now,

$$\begin{aligned} A^3 &= A^2.A \\ &= 2 \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \end{aligned}$$

$$A^3 = 2 \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$A^3 = 4 \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$A^3 = 4A$$

 $\therefore$ statement 2 is correct

Sol.44. (b)

Zero matrix or Null matrix

Sol.45. (d)

$$\begin{aligned} \text{We have, } [x \ y \ z] &= \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \end{aligned}$$

$$= [xa + yh + zg \quad xh + yb + zf \quad xg + yf + zc]$$

Sol.46. (c)

$$\text{Given, } m = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } n = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\therefore m \cos \theta = \begin{bmatrix} \cos \theta & 0 \\ 0 & \sin \theta \end{bmatrix}$$

$$\text{and } n \sin \theta = \begin{bmatrix} 0 & \sin \theta \\ -\sin \theta & 0 \end{bmatrix}$$

$$\therefore m \cos \theta - n \sin \theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{aligned} \therefore \text{Determinant of } (m \cos \theta - n \sin \theta) \\ &= \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = 1 \end{aligned}$$

Sol.47. (a)

$$f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore f(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and}$$

$$f(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore f(\theta) \cdot f(\phi) =$$

$$\begin{aligned} &\begin{bmatrix} \cos \theta \cdot \cos \phi - \sin \theta \cdot \sin \phi + 0 & \sin \theta \cdot \cos \phi + \sin \theta \sin \phi + 0 & 0 \\ -\cos \theta \cdot \sin \phi - \sin \theta \cdot \cos \phi + 0 & 0 + 0 + 0 & 0 \\ -\cos \theta \cdot \sin \phi + \sin \theta \cdot \cos \phi + 0 & 0 + 0 + 0 & 0 + 0 + 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & 0 \\ \sin(\theta+\phi) & \cos(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix} = f(\theta+\phi) \end{aligned}$$

 Now, determinant of $f(\theta) \times f(\phi)$

$$\begin{bmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & 0 \\ \sin(\theta+\phi) & \cos(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \cos^2(\theta+\phi) + \sin^2(\theta+\phi) = 1$$

$$\text{Since, determinant of } f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \cos^2 x + \sin^2 x = 1$$

 $\therefore$ Determinant of $f(x)$ is an even number.

Sol.48. (d)

We have,

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix}$$

 $\therefore AB$

$$= \begin{bmatrix} 1 \times 2 + (-1) \times (-1) & 1 \times 3 + (-2) \\ 2 \times 2 + 3 \times (-1) & 2 \times 3 + 3 \times (-2) \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 8 & 7 \\ -5 & -5 \end{bmatrix}$$

 Clearly, $AB \neq BA$

 Now, $AB(A^{-1}B^{-1}) = AB(BA)^{-1}$

$$[\because (AB)^{-1} = B^{-1}A^{-1}]$$

 $AB(BA)^{-1}$ is a unit matrix if and only if $AB = BA$ but $AB \neq BA$.

 So, $AB(A^{-1}B^{-1})$ is not a unit matrix

Since, $AB \neq BA$, therefore $(AB)^{-1} B^{-1}$

Sol.49. (b)

$$A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$\therefore A \cdot A^T = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \alpha + \sin^2 \alpha & -\cos \alpha \sin \alpha + \cos \alpha \sin \alpha \\ -\sin \alpha \cos \alpha + \cos \alpha \sin \alpha & \cos^2 \alpha + \sin^2 \alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Sol.50. (d)

It is given that $AB = C$

$$\begin{bmatrix} x+y & y \\ x & x-y \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

$$3x + 3y - 2y = 4 \text{ or } 3x + y = 4 \quad \dots(i)$$

$$\text{and } 3x - 2x + 2y = -2 \text{ or } x + 2y = -2 \quad \dots(ii)$$

solving equations (i) and (ii), we get $x = 2$ and $y = -2$

$$\therefore A^2 =$$

$$\begin{bmatrix} 0 & -2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 0 & -2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 0-4 & 0-8 \\ 0+8 & -4+16 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & -8 \\ 8 & 12 \end{bmatrix}$$

Sol.51. (a)

$$A^2 = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 \theta - \sin^2 \theta & \cos \theta \sin \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta - \cos \theta \sin \theta & -\sin^2 \theta + \cos^2 \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$\therefore A^3 =$$

$$\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$$

Sol.52. (b)

Order of the resultant matrix

$$= (1 \times 3) \text{ by } (3 \times 3) \text{ by } (3 \times 1) = (1 \times 1)$$

Sol.53. (a)

We know that,

$$A^4 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

this is an identity matrix.

Sol.54. (a)

$$A = \begin{bmatrix} 4i-6 & 10i \\ 14i & 6+4i \end{bmatrix}$$

$$\Rightarrow kA = \frac{1}{2i} \begin{bmatrix} 4i-6 & 10i \\ 14i & 6+4i \end{bmatrix} = \begin{bmatrix} 2-\frac{3}{i} & 5 \\ 7 & 3+\frac{3}{i}+2 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\frac{3i}{i^2} & 5 \\ 7 & \frac{3i}{i^2}+2 \end{bmatrix} = \begin{bmatrix} 2+3i & 5 \\ 7 & 2-3i \end{bmatrix}$$

Sol.55. (c)

Since $\text{adj } A^T - (\text{adj } A)^T$

$\therefore \text{adj } A^T - (\text{adj } A)^T = \text{null matrix of same order as that of } A.$

Sol.56. (a)

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\therefore |A| = a(bc) = abc$$

$$\text{Cofactor matrix of } A \text{ (say } B) = \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$$\Rightarrow \text{adj. } A = B^T = \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$$\therefore A^{-1}$$

$$\frac{\text{adj. } A}{|A|} = \frac{1}{abc} \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$$= \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$$

Sol.57. (b)

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

$$\text{Cofactor matrix of } A = \begin{bmatrix} +1 & -2 & +6 \\ +6 & +1 & -3 \\ -2 & +4 & +1 \end{bmatrix}$$

$$\text{adjoint } A = \begin{bmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{bmatrix}$$

Sol.58. (b)

$$A = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = \begin{bmatrix} 4+4 & -4-4 \\ -4-4 & 4+4 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -8 \\ -8 & 8 \end{bmatrix} = -4 \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = -4A$$

Sol.59. (c)

$$A_{x,x+5} \text{ and } B_{y,11-y}$$

If AB exists then $x+5 = y$ or $x-y = -5$

And if BA exists then $11-y = x$ or $x+y = 11$

Solving equations, (i) and (ii), we get $x = 3$ and $y = 8$.

Sol.60. (b)

If α and β are the root of the equation

$$1 + x + x^2 = 0, \text{ then } \alpha = \omega \text{ and } \beta = \omega^2$$

$$\therefore \begin{bmatrix} 1 & \beta \\ \alpha & \alpha \end{bmatrix} \begin{bmatrix} \alpha & \beta \\ 1 & \beta \end{bmatrix} = \begin{bmatrix} 1 & \omega^2 \\ \omega & \omega \end{bmatrix} \begin{bmatrix} \omega & \omega^2 \\ 1 & \omega^2 \end{bmatrix}$$

$$= \begin{bmatrix} \omega + \omega^2 & \omega^2 + \omega^4 \\ \omega^2 + \omega & \omega^3 + \omega^3 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & 2 \end{bmatrix}$$

Sol.61. (a)

$$A^2 - I_2 = kA$$

$$\Rightarrow \begin{bmatrix} 5 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} k & 2k \\ 2k & 3k \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & 8 \\ 8 & 12 \end{bmatrix} = \begin{bmatrix} k & 2k \\ 2k & 3k \end{bmatrix}$$

$$\Rightarrow k = 4$$

Sol.62. (a)

$$(A)_{(2 \times 3)} \times (B)_{(3 \times 5)} = (AB)_{2 \times 5}$$

$\therefore B$ must be 3×5 matrix

Sol.63. (a)

In this case, $A^{-1} = \text{adj } A = (\text{co-factor } A)^T$

$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol.64. (b)

Matrix product is commutative if both are diagonal matrices of same order.

$$\Rightarrow A^2 - B^2 = (A+B)(A-B) \text{ is not true}$$

$$\text{Next, } (A-I)(A+I) = 0$$

$$\Rightarrow A^2 + AI - IA - I^2 = 0 (\because AI = IA)$$

$$\Rightarrow A^2 = I^2 \text{ is correct.}$$

Sol.65. (b)

$B = \text{adj } A$, I Identity matrix, $|A| = k$

$$AB = A(\text{adj } A) = |A|I = kI$$

Sol.66. (c)

By property, statement I and III are correct.

Sol.67. (b)

$$\begin{vmatrix} 2 & 4 \\ -8 & x \end{vmatrix} = 0 \Rightarrow x = -16$$

Sol.68. (a)

$$(AB)^{-1} = B^{-1}A^{-1}$$

Sol.69. (a)

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

Sol.70. (b)

If $A^{-1} = A^T$, then A is orthogonal matrix.

Sol.71. (b)

Inverse of an identity matrix is an identity matrix

Sol.72. (a)

$$B = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 4 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\text{Minor} = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\text{cofactor} = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & -1 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\text{Adjoint} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & -1 & 8 \end{bmatrix}$$

Sol.73. (b)

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^2 = I \text{ (Involutory matrix)}$$

Sol.74. (a)

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}$$

$$A^3 - 2A^2 = \begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix} - 2 \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Sol.75. (c)

A is 3×2 order matrix and BA is 2×2 order matrix.

so AB exists but BA not exists.

Sol.76. (c)

A is 3×3 and B is 3×1 so order of AB will be 3×1

$$AB = \begin{bmatrix} ax+hy+gz \\ hx+by+fz \\ gx+fy+cz \end{bmatrix}$$

Sol.77. (c)

Order of A is 3×5

and order of B is 5×3

Then order of AB will be 3×3 and BA will be 5×5

Sol.78. (c)

$$A = \begin{bmatrix} 1-i & i \\ -i & 1-i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1+i & -i \\ i & 1+i \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} 1+i & i \\ -i & 1+i \end{bmatrix}$$

$$(\bar{A})^T + A = \begin{bmatrix} 1+i & i \\ -i & 1+i \end{bmatrix} + \begin{bmatrix} 1-i & i \\ -i & 1-i \end{bmatrix}$$

$$(\bar{A})^T + A = B = \begin{bmatrix} 2 & 2i \\ -2i & 2 \end{bmatrix}$$

this matrix is hermit ion matrix because

$$(\bar{B})^T = B$$

Sol.79. (d)

for singular matrix, determinant of this matrix should be zero.

$$-k(k-5k)+4(-k^2)=0$$

$$-5k^2+5k^2=0$$

for every value of k this equation is singular.

Sol.80. (b)

$$A = \begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix}, B = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$AB = C$$

$$\begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 2x+y \\ 3x+y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$2x+y=3 \dots\dots\dots(i)$$

$$3x+y=2 \dots\dots\dots(ii)$$

by solving above two equations

$$x=-1 \text{ and } y=5$$

$$\text{then } A = \begin{bmatrix} 4 & 5 \\ -2 & -6 \end{bmatrix} \text{ and } |A| = -14$$

Sol.81. (a)

$$I. A(\text{adj}A) = (\text{adj}A)$$

always correct for a square matrix

$$II. |\text{adj} A| = |A|$$

incorrect for order 3 because for order 3

$$|\text{adj} A| = |A|^2$$

Sol.82. (d)

order of A and B need not be same.

for example $A_{2 \times 3} B_{3 \times 2} = (AB)_{2 \times 2}$

Sol.83. (c)

prime numbers = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29
from these 10 prime numbers 4 matrices of order (1×10) , (2×5) , (5×2) , (10×1) are possible.

Sol.84. (a)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A \in \mathbb{N}$$

$$\Rightarrow A \cdot A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix}$$

$$\text{So } A^{100} = \begin{bmatrix} 1 & 100a \\ 0 & 1 \end{bmatrix}; A^{50} = \begin{bmatrix} 1 & 50a \\ 0 & 1 \end{bmatrix};$$

$$A^{25} = \begin{bmatrix} 1 & 25a \\ 0 & 1 \end{bmatrix}$$

$$\text{So } A^{100} - A^{50} - 2A^{25}$$

$$\begin{bmatrix} 1 & 100a \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 50a \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 25a \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 100a \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 50a \\ 0 & 1 \end{bmatrix} - 2 \begin{bmatrix} 2 & 25a \\ 0 & 2 \end{bmatrix}$$

$$= -2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ or } -2I$$

Sol.85. (d)

Here we have to form distinct matrices with four entries taken from $\{1, 2\}$

So matrices formed may be of

$$4 \times 1 \text{ or } 1 \times 4 \text{ or } 2 \times 2$$

So we have 3 types of matrices and each on each position either the value is 2 or 1 so there are 2 ways for each position and four entries.

$$\text{So ans is } = 3 \times [2 \times 2 \times 2 \times 2] = 48$$

Sol.86. (c)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Here (1) inverse e of A does not exist is true because all entries are same

$$\text{So } |A| = 0$$

$$\Rightarrow A^{-1} \text{ ie } A^{-1} = \frac{\text{adj}A}{|A|} \text{ become infinite}$$

So (1) is correct

$$2. \text{When } A^2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \text{ or } 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\text{If } A^3 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

This gives yous $3A = A^3$

So, second statement ie $A^3 = A$ is incorrect and 3rd statement is correct

Option is correct.

Sol.87. (a)

$$A = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

$$\text{Adj. } A = \begin{bmatrix} -1/2 & -1 \\ -3/2 & -2 \end{bmatrix}$$

$$|A| = -1/2$$

$$A^{-1} = \frac{\text{adj}A}{|A|} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Sol.88. (d)

$$A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$(mI + nA) =$$

$$\left(\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} + \begin{bmatrix} 0 & 2n \\ -2n & 0 \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} m & 2n \\ -2n & m \end{bmatrix}^2 = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} m^2 - 4n^2 & 4mn \\ -4mn & m^2 - 4n^2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$m^2 - 4n^2 = 0$$

$$m^2 = 4n^2$$

$$m = 2n$$

$$4mn = 2$$

$$4(2n)n = 2$$

$$n = \frac{1}{2}$$

$$\text{then } m = 1$$

$$m + n = 1 + \frac{1}{2} = \frac{3}{2}$$

Sol.89. (c)

$$1. CA = \begin{bmatrix} m \\ -m \end{bmatrix} \begin{bmatrix} m & n \end{bmatrix} = \begin{bmatrix} m^2 & mn \\ -m^2 & -mn \end{bmatrix}$$

$$CB = \begin{bmatrix} m \\ -m \end{bmatrix} \begin{bmatrix} -n & -m \end{bmatrix} = \begin{bmatrix} -mn & -m^2 \\ mn & +m^2 \end{bmatrix}$$

$$CA \neq CB$$

Statement 1 is incorrect

$$2. AC = \begin{bmatrix} m & n \end{bmatrix} \begin{bmatrix} m \\ n \end{bmatrix} = \begin{bmatrix} m^2 - mn \end{bmatrix}$$

$$BC = \begin{bmatrix} -n & -m \end{bmatrix} \begin{bmatrix} n \\ -m \end{bmatrix} = \begin{bmatrix} -mn + m^2 \end{bmatrix}$$

Statement 2 is also correct.

3. $C(A+B) = CA + CB$ (distribution law always correct)

Sol.90. (d)

$$A(\text{adj.}A) = |A|I_n$$

$$A = \begin{bmatrix} 2\sin\theta & \cos\theta & 0 \\ -2\cos\theta & \sin\theta & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$|A| = 2\sin^2\theta + 2\cos^2\theta = 2$$

$$\text{then } A(\text{adj.}A) = 2I$$

Sol.91. (d)

For singular matrix $|A|=0$

$$\begin{vmatrix} 2\cos 2\theta & 2\cos 2\theta & 6 \\ 1-2\sin^2\theta & 2\cos^2\theta-1 & 3 \\ k & 2k & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 2\cos 2\theta & 2\cos 2\theta & 6 \\ \cos 2\theta & \cos 2\theta & 3 \\ k & 2k & 1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 - 2R_2$$

$$= \begin{vmatrix} 0 & 0 & 0 \\ \cos 2\theta & \cos 2\theta & 3 \\ k & 2k & 1 \end{vmatrix} = 0$$

Value of determinant is not depending upon the value of k.

Sol.92. (b)

Given than $B = \text{adj}(A)$

$$1. A(\text{adj} A) = (\text{adj} A) A \text{ correct}$$

$$2. AB = A(\text{adj} A) = |A| I_n = \text{scalar matrix correct.}$$

1 and 2 only correct.

Sol.93. (b)

$$1. AB = 0$$

then $A = 0$ or $B = 0$ or $A \neq 0$ & $B \neq 0$

Incorrect statement

2. $AB = I$ then $A = A^{-1}$

$A^{-1}A = AA^{-1} = I$

Correct statement

Sol.94. (d)

$B = A^T$

$C = A + A^T$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$|C| = 8$

Sol.95. (c)

$AB = A$ & $BA = B$ then $A^2 = A$ & $B^2 = B$

Both statements are correct.

Sol.96. (a)

$$P = \frac{1}{2}(A + A^T)$$

$$= \frac{1}{2} \left[\begin{pmatrix} 0 & \sin^2 \theta & \cos^2 \theta \\ \cos^2 \theta & 0 & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta & 0 \end{pmatrix} + \begin{pmatrix} 0 & \cos^2 \theta & \sin^2 \theta \\ \sin^2 \theta & 0 & \cos^2 \theta \\ \cos^2 \theta & \sin^2 \theta & 0 \end{pmatrix} \right]$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

Sol.97. (d)

$$Q = \frac{1}{2}(A - A^T)$$

$$= \frac{1}{2} \left[\begin{pmatrix} 0 & \sin^2 \theta & \cos^2 \theta \\ \cos^2 \theta & 0 & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta & 0 \end{pmatrix} - \begin{pmatrix} 0 & \cos^2 \theta & \sin^2 \theta \\ \sin^2 \theta & 0 & \cos^2 \theta \\ \cos^2 \theta & \sin^2 \theta & 0 \end{pmatrix} \right]$$

$$= \frac{1}{2} \begin{pmatrix} 0 & \sin^2 \theta - \cos^2 \theta & \cos^2 \theta - \sin^2 \theta \\ \cos^2 \theta - \sin^2 \theta & 0 & \sin^2 \theta - \cos^2 \theta \\ \sin^2 \theta - \cos^2 \theta & \cos^2 \theta - \sin^2 \theta & 0 \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 0 & -\cos 2\theta & \cos 2\theta \\ \cos 2\theta & 0 & -\cos 2\theta \\ -\cos 2\theta & \cos 2\theta & 0 \end{pmatrix}$$

$$= \frac{1}{2} \cos 2\theta \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}$$

Sol.98. (a)

$$A = \begin{pmatrix} 0 & \sin^2 \theta & \cos^2 \theta \\ \cos^2 \theta & 0 & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta & 0 \end{pmatrix}$$

$|A| = \sin^6 \theta + \cos^6 \theta$.

And its min value exists at 45°

So $\text{Min}|A| = \frac{1}{4}$

Sol.99. (d)

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & \sin \theta & -\cos \theta \end{pmatrix}$$

$$\text{adj}A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\cos \theta & -\sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & \sin \theta & -\cos \theta \end{pmatrix}$$

, then which of the following are correct?

1. $A + \text{adj}A$ is a null matrix (correct)

2. $A^{-1} + \text{adj}A$ is null matrix (correct)

3. $A - A^{-1}$ is a null matrix (correct)

Sol.100. (d)

If X is a matrix of order 3×3 , Y is a matrix of order 2×3 and Z is a matrix of order 3×2 , then which of the following are correct?

1. (ZY) is a 3×3 matrix,

then $(ZY)X$ is a 3×3 matrix so matrix having 9 entries

2. (XZ) is a 3×2 matrix,

Then $Y(XZ)$ is a 2×2 matrix so having 4 entries

3. $X(YZ)$ is not defined because number of columns in X matrix is not equal to number of rows in YZ matrix

All above statements are correct

Sol.101. (c)

A and B be symmetric matrices

$(AB - BA)' = (AB)' - (BA)'$

$(AB)' - (BA)' = B'A' - A'B' = BA - AB$

$(AB - BA)' = BA - AB =$

$(AB - BA)' = -(AB - BA)$

So $(AB - BA)$ is a skew symmetric matrix and

for any skew symmetric matrix below given statements are always correct

1. Its diagonal entries are equal but nonzero

2. The sum of its non-diagonal entries is zero

Sol.102. (c)

1. $AB = AC$

$\Rightarrow A^{-1}AB = A^{-1}AC$

$\Rightarrow B = C$

2. If $BX = CX$

If X is a column matrix then

$B = C$

Both statements are correct

Sol.103. (b)

X_1 and X_2 be its two distinct solutions of $AX = B$

So $AX_1 = B$ (i)

And $AX_2 = B$ (ii)

$aX_1 + bX_2$ is a solution of $AX = B$

so $A(aX_1 + bX_2) = B$

by (i) and (ii)

$aB + bB = B$

$a + b = 1$

Sol.104. (a)

$$AA' = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

$(I + AA') =$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 6 & 10 \end{bmatrix}$$

$|I + AA'| = 28 - 4 - 9 = 15$

Sol.105. (a)

$$A = \begin{bmatrix} 0 & 3 & 4 \\ -3 & 0 & 5 \\ -4 & -5 & 0 \end{bmatrix},$$

$\Rightarrow (A^2)' = (AA)' = A'A'$

[Given matrix is a skew symmetric

So $A' = -A]$

$= (-A)(-A) = A^2$

So A^2 is a symmetric matrix.

Determinant of an odd ordered skew symmetric matrix is zero

So $|A| = 0$, then $|A^n| = 0$

Sol.106. (b)

$$\text{If } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \text{ then which one of the}$$

following statements are correct?

1. $|A| \neq 0$, so A and A^n is not a singular matrix

2. if we multiply diagonal matrix to any other diagonal matrix then result will be a diagonal matrix.

3. Diagonal matrix is always a symmetric matrix

Only statement 2 and 3 are correct.

Sol.107. (d)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 3 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow |A| = 5$

$\Rightarrow |\text{adj}(\text{adj}A)| = |A|^{(n-1)^2} = 5^4 = 625$

Sol.108. (a)

$$\text{If } A = A^2 = A^3 = A^n = \text{unit matrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$23A^2 - 19A^2 - 4A$

$$= 23 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 19 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Sol.109. (d)

Number of matrices of order 2×2 are $4 \times 4 \times 4 \times 4 = 256$

Number of matrices of order 1×4 are $4 \times 4 \times 4 \times 4 = 256$

Number of matrices of order 4×1 are $4 \times 4 \times 4 \times 4 = 256$

Total matrices = 768

Sol.110. (c)

$$A = \begin{bmatrix} \sin 2\theta & 2\sin^2 \theta - 1 & 0 \\ \cos 2\theta & 2\sin \theta \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} \sin 2\theta & -\cos 2\theta & 0 \\ \cos 2\theta & \sin 2\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\det(A) = 1$

so $A^{-1} = \text{adj}(A)$

diagonals are not zero so matrix is not a skew symmetric matrix.

and $A^{-1} = A^T$

statement 1 and 3 are correct.

Sol.111. (b)

$$\text{adj}(AB) = (\text{adj}B)(\text{adj}A)$$

so statement 1 and 2 are incorrect.

$$(A)\text{adj}(A) = |A|I_n$$

Statement 3 is correct.

Sol.112. (a)

from properties of adjoint

$$A(\text{adj}A^T) = A(\text{adj}A)^T$$

statement 1 is correct.

$$A^2 = A$$

$$A(A - I) = 0$$

$$|A| = 0 \text{ or } |A| = 1$$

matrix is non singular given in question so $|A| = 1$, and orthogonal matrix of $|A| = 1$ is always a identity matrix.

statement 2 is correct.

$$A^3 = A$$

$$A(A^2 - I) = 0$$

$$|A| = 0 \text{ or } |A| = +1 \text{ or } -1$$

matrix is non singular given in question so $|A| = +1 \text{ or } -1$

so always unit matrix is not correct.

statement 3 is wrong.

Sol.113. (a)

diagonals of skew symmetric matrix is always zero. so statement 1 and 2 are correct.

$$\text{orthogonal matrix } A^{-1} = A^T \text{ or } AA^T = I$$

skew symmetric matrices can never be orthogonal matrix.

Sol.114. (d)

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

for above matrix $|A| = 1$

$$A(\text{adj}A) = |A| I_n$$

$$A(\text{adj}A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol.115. (a)

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

for above matrix $|A| = 1$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \text{adj} A = \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & -3 \end{bmatrix}$$

Sol.116. (c)

$$P = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

and

$$Q = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix}$$

$$PQ = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$QP = \begin{bmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{bmatrix} \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$PQ = QP = \text{Null matrix}$$

statement I and III are correct.

Sol.117. (c)

If X is a matrix of order 3×3 , Y is a matrix of order 2×3 and Z is a matrix of order 3×2 , then which of the following are correct?

1. (ZY) is a 3×3 matrix,

then (ZY)X is a 3×3 matrix

2. (XZ) is a 3×2 matrix,

then Y(XZ) is a 2×2 matrix

3. X(YZ) is not defined because number of columns in X matrix is not equal to number of rows in YZ matrix

All above statements are correct

Sol.118. (c)

$$AB = 0$$

If any one matrix is non-singular than second matrix should be a null matrix.

Sol.119. (d)

$$\begin{bmatrix} x+4+7 & 2x+5+8 & 3x+6+9 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = \begin{bmatrix} 45 \end{bmatrix}$$

$$\begin{bmatrix} x+11+2x+13+3x^2+15x \end{bmatrix} = \begin{bmatrix} 45 \end{bmatrix}$$

$$3x^2 + 18x - 21 = 0$$

$$x^2 + 6x - 7 = 0$$

$$(x+7)(x-1) = 0$$

$$x = 1, -7$$

Sol.120. (b)

$$A = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix}$$

If A is orthogonal than $A.A^T = I$

$$AA^T = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix} \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix} = I$$

$$= \begin{bmatrix} x^2+y^2+z^2 & xy+yz+zx & xy+yz+zx \\ xy+yz+zx & x^2+y^2+z^2 & xy+yz+zx \\ xy+yz+zx & xy+yz+zx & x^2+y^2+z^2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x^2+y^2+z^2 = 1$$

Sol.121. (c)

$$|M^{-1}| = \frac{1}{|M|}$$

2nd property is wrong, others are correct

Sol.122. (d)

$$f(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$f(\pi) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\{f(\pi)\}^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Sol.123. (d)

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$A^2 - 4A = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= 5I_3$$

Sol.124. (b)

$$A = \begin{bmatrix} y & z & x \\ z & x & y \\ x & y & z \end{bmatrix}$$

If A is orthogonal than $A.A^T = I$

$$A^2 = I$$

Sol.125. (a)

$$\text{I. } (AB)^{-1} = B^{-1}A^{-1}$$

$$\text{II. } (BA)(AB)^{-1} = (BA)(B)^{-1}(A)^{-1}$$

$$\text{III. } (AB)^T = B^T A^T$$

So all are incorrect.